Mohammad (Mahdi) Abootorabi

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For comprehensive insights into each section, kindly refer to my Homepage.

Education

The University of British Columbia *Vancouver, Canada*

Master of Applied Science in Electrical and Computer Engineering *Sep. 2025 - Aug. 2027 (expected)*

- Overall **GPA**: 96.5% (Average: **A+**)
- Under the supervision of **Professor Purang Abolmaesumi**
- **Notable Courses**: Machine Teaching, Adaptation & Adaptive Computation, Interpretability and Explainability, NLP Commonsense

Sharif University of Technology (SUT) *Tehran, Iran*

Bachelor of Science in Computer Engineering *Sep. 2019 - Feb. 2024*

- Overall **GPA**: 3.93/4.00 (Average: **19.38/20**) , Major **GPA**: 3.97/4.00 (Average: **19.73/20**)
- **Notable Courses**: Natural Language Processing, Machine Learning, Artificial Intelligence, Linear Algebra, Modern Information Retrieval

Research Interests

- Multimodal Learning, Natural Language Processing, Interpretability & explainability, Representation Learning, AI for Healthcare

Selected Publications

- V. Suresh*, **M. M. Abootorabi***, M. Salman, M. H. Mughal, C. Theobalt, A. Ram, J. Steimle, and V. Demberg. Semantic Motion Anchors: Bridging Motion and Meaning in Co-Speech Gestures. *Submitted to Neurips 2026*. Preprint
- **M. M. Abootorabi**, A. Saghafian, A. Bazshoushtari, H. Rezaei, E. Hwang, V. Shwartz, P. Mousavi, and P. Abolmaesumi. Steering Geometry: Validating Human Value Geometry in LLM Steering Space. *Submitted to EMNLP 2026*. Preprint
- **M. M. Abootorabi**, P. Mousavi, P. Abolmaesumi, and E. Shelhamer. ProtoTTA: Prototype-Guided Test-Time Adaptation. *Third Workshop on Test-Time Updates (Main Track) at ICLR 2026*, Submitting to TMLR, April 2026. OpenReview Github
- **M. M. Abootorabi**, S. Namazi, A. S. Boroujeni, L. Wang, O. Dzikunu, P. F. R. Wilson, Z. Guo, B. Wodlinger, P. Mousavi, and P. Abolmaesumi. Learning Where to Look: A Reinforcement Learning Framework for Robust Micro-Ultrasound Prostate Cancer Detection. *Early Accepted at MICCAI 2026 (top 9%)*. Preprint
- **M. M. Abootorabi***, O. Ghahroodi*, A. Madkoor, M. Nouri, D. Dastgheib, and E. Asgari. Almieyar-Oryx-BloomBench: A Bilingual Multimodal Benchmark for Cognitively Informed Evaluation of Vision-Language Models. *ACL 2026 Findings*, July 2026. Preprint
- L. Wang, **M. M. Abootorabi**, A. Saadat, V. Wu, N. Hashemi, S. Sojoudi, X. He, L. Sigal, C. Luong, T. Tsang, and P. Abolmaesumi. EchoAFR: Echocardiography multi-video report generation through anatomical and functional aware hierarchical Q-former. *Submitted to MICCAI 2026*. Preprint
- O. K. Dzikunu, **M. M. Abootorabi**, M. Harmanani, P. F. R. Wilson, E. Willis, F. Luger, A. Kinnaird, B. Wodlinger, P. Mousavi, and P. Abolmaesumi. Learning Prostate Anatomy at Test Time for Cancer Detection in Micro-Ultrasound. *Submitted to MLHC 2026*. Preprint
- E. Willis*, T. Elghareb*, P. F.R. Wilson, M.N.N. To, **M. M. Abootorabi**, A. Jamzad, B. Wodlinger, P. Mousavi, and P. Abolmaesumi. GUIDE-US: Grade-Informed Unpaired Distillation of Encoder Knowledge from Histopathology to Micro-UltraSound. *IPCAI 2026*. arXiv
- **M. M. Abootorabi**, A. Zobeiri, M. Dehghani, M. Mohammadkhani, B. Mohammadi, O. Ghahroodi, M. S. Baghshah, and E. Asgari. Ask in Any Modality: A Comprehensive Survey on Multimodal Retrieval-Augmented Generation. *ACL 2025 Findings*, July 2025. +115 Citations and +500 GitHub Stars. ACL Anthology Github Website
- **M. M. Abootorabi** and E. Asgari. CLASP: Contrastive Language-Speech Pretraining for Multilingual Multimodal Information Retrieval. *ECIR 2025*, April 2025. arXiv ACM Digital Library Website Models Newly Introduced Dataset Code
- **M. M. Abootorabi**, O. Ghahroodi, P. S. Zahraei, H. Behzadasl, A. Mirrokni, M. Salimipanah, A. Rasouli, B. Behzadipour, S. Azarnoush, B. Maleki, E. Sadraiye, K. K. Feriz, M. T. Nahad, A. Moghadasi, A. E. Abianeh, N. Nazar, H. R. Rabiee, M. S. Baghshah, M. Ahmadi, and E. Asgari. Generative AI for Character Animation: A Comprehensive Survey of Techniques, Applications, and Future Directions. May 2025. 23 Citations arXiv Github Website
- **M. M. Abootorabi**, N. Ghazizadeh, A. Dalili, A. Ghahramani, M. Dehghani, and E. Asgari. AIMA at SemEval-2024 Task 10: History-Based Emotion Recognition in Hindi-English Code-Mixed Conversations. *SemEval at NAACL 2024*, June 2024. ACL Anthology
- S. Elahimanesh, M. Mohammadkhani*, S. Z. Movahed*, **M. M. Abootorabi***, S. Salehi, and A. Edalat. Structure Matters: Evaluating Multi-Agent Orchestration in Generative Therapeutic Chatbots. *ACM CUI 2026*. arXiv

Skills

Programming Languages	**Python**, **Java**, C, C++, Android, HTML, JavaScript
Frameworks	**NumPy**, **PyTorch**, **Transformers**, **Scikit-Learn**, **W&B**, PyTorch Lightning, NLTK, SciPy, Pandas, Django, REST
Databases	**PostgreSQL**, **Redis**, MongoDB
Operating Systems	**Linux (Ubuntu)**, Microsoft Windows, macOS, WSL
Other Technologies	**Git**, **Celery**, RabbitMQ, Jupyter Notebook, LaTeX, Docker, Trello, Notion, Prompt Engineering
Soft Skills	**Teamwork**, **Self-Learning**, **Leadership**, Problem Solving, Creativity, Flexibility, Mentorship, Willingness to Learn

Professional Experience

The University of British Columbia and Vector Institute *Sep. 2025 – Present*

Research Assistant *Supervisors: Prof. Purang Abolmaesumi & Prof. Parvin Mousavi*

- Developing novel **multimodal learning** and **reinforcement learning** methods to improve prostate cancer detection using ultrasound images.

Teaching Assistant *Sep. 2025 – Present*

- Served as a Teaching Assistant for multiple courses, delivering mini-lectures, leading lab sessions, mentoring project teams, and grading.

Saarland University *June. 2025 – Sep. 2025*

Research Assistant *Supervisor: Prof. Vera Demberg*

- Designing **contrastive learning-based methods** to enhance **semantic gesture understanding and representation** from video, with a publication in preparation.

Qatar Computing Research Institute (QCRI) — *July. 2024 – Sep. 2025*

REMOTE RESEARCH ASSISTANT — *Supervisor: Dr. Ehsaneddin Asgari*

- Author surveys on **multimodal RAG** and generative AI for animated characters, highlighting gaps and future directions.
- Advance Arabic vision-language models beyond state-of-the-art and **introduce a benchmark** for practical VLM evaluation.

Imperial College London — *Sep. 2024 – Sep. 2025*

REMOTE RESEARCH ASSISTANT (VOLUNTARY, UNPAID) — *Supervisor: Prof. Abbas Edalat*

- Design an **LLM-based Persian therapist Agent** incorporating the Self-Attachment Technique (SAT), achieving superior performance over baseline LLMs.

University of New South Wales — *June. 2023 – Sep. 2023*

REMOTE RESEARCH ASSISTANT (VOLUNTARY, UNPAID, SUMMER INTERNSHIP) — *Supervisor: Dr. Imran Razzak*

- Design an **RL-based model** to automate and enhance medical report generation for Fundus Fluorescein Angiography images.

Yektanet Technology Company — *Oct. 2020 - Oct. 2021*

SOFTWARE ENGINEER

- Develop **innovative algorithms** that improve user attraction to online ads, driving higher income.
- Develop **an ATS system** from scratch to automate tasks such as scheduling meetings and writing Trello comments, streamlining the HR process.
- Implement features in a large-scale service and **mentor fresh software engineers** to support their professional growth.
- Handle site reliability engineering tasks, including **server maintenance** and performance monitoring.
- **Design efficient data storage methods** and work with databases and architectures such as Redis master-slave to manage high request volumes.

Sharif University of Technology — *Jan. 2023 – Feb. 2024*

RESEARCH ASSISTANT (VOLUNTARY, UNPAID) — *Supervisor: Dr. Ehsaneddin Asgari*

- Design a novel **multimodal contrastive learning**-based semantic speech encoder that outperforms ASR-based retrieval pipelines in speed, accuracy (many scenarios), and efficiency.

TEACHING ASSISTANT — *Sep. 2020 - June 2024*

- Served as **Head TA** for Compiler Design ($\times 5$) and TA for Artificial Intelligence ($\times 3$), Modern Information Retrieval ($\times 3$), and Linear Algebra ($\times 3$).
- **Led and organized TA teams**, coordinating grading workflows for mid-term and final exams to ensure consistency and efficiency.
- Designed comprehensive **assignments, practical projects, and exam questions**, while also contributing to curriculum improvement by **designing and revising lecture slides**.

Remarkable Academic Projects

Autonomous Vision-Powered Travel Agent *Source Code* — *Spring 2025*

- Developed an autonomous travel-planning agent that processes natural language queries and navigates booking websites **without relying on HTML.**
- Leveraged **multimodal integration** of Computer Vision and LLMs to interpret screenshots and present flight options via an interactive front-end.

RAG-based Educational Assistant Chatbot *Source Code* — *Winter 2025*

- Built an **end-to-end educational chatbot** using a RAG architecture to deliver reliable, country-specific student guidance.
- Integrated Vue, Node.js, and FastAPI with **Sentence Transformers and LLMs** to ground responses in verified documents; deployed live for personalized advice based on user context.

Health Retrieval Search Engine *Source Code* — *Spring 2022*

- Designed an **advanced search engine** for **efficient retrieval** and organization of health-related content.
- Implemented TF-IDF, FastText, Elasticsearch, and Transformer models using modern techniques with a deployable user-friendly interface.

Bias detection & De-biasing in Language Models *Source Code* — *Winter 2022*

- Detected and **mitigated gender and ethnic biases** in multilingual language models, including low-resource languages.
- Developed **de-biasing techniques** specifically tailored for **Persian language models.**

Personal Information Hider *Source Code* — *Spring 2022*

- Developed a Python package to **enhance privacy** by concealing personal information in text files using POS tagging and regex-based methods.

Resume Parser *Source Code* — *Winter 2022*

- Created a Python package for **automated resume parsing**, extracting structured information from Persian CVs to support Applicant Tracking Systems.

Honors & Awards

2026	**Selected for the Graduate Support Initiative (GSI) Award and the MUSIC Program Graduate Student Award**, to receive UBC-funded support recognizing research excellence and graduate academic achievement.	*Canada*
2025	**Selected for the Konrad Zuse School (ELIZA) Scholarship**, to join a graduate school of excellence connecting German units of the European ELLIS network and Germany's leading experts in machine learning-driven AI.	*Germany*
2024	**Ranked in the top 9%**, of Computer Engineering students based on GPA at Sharif University of Technology. (**16** out of 190)	*Iran*
2019	**Ranked 10th**, among nearly 165,000 participants in the nationwide university entrance exam (Konkour) in the Mathematics Branch, securing a fully funded B.Sc. (tuition fee waiver) period in Iran.	*Iran*
2013	**Admitted to Allameh Helli Middle School**, among nearly 80,000 students under the supervision of the National Organization for Developing Exceptional Talents (NODET).	*Iran*

Academic Service & Volunteer Activities

Reviewer, ACM Transactions on Intelligent Systems and Technology and MICCAI 2026 — *Fall 2025*

Volunteer Staff at ACL 2025 Conference — *Summer 2025*

- Assisted in organizing the conference by coordinating presentations and poster sessions, and supporting attendee registration.

Technical Content Creation — *Fall 2022 – Spring 2023*

- Expanded Persian Wikipedia's Machine Learning resources by creating a page on **Knowledge Distillation** (link) and significantly enhancing the page on **Autoencoders** (link), making advanced ML concepts more accessible to Persian readers.
- Authored educational content on **Test-driven Development** by creating posts for the Persian Medium and enhancing its Persian Wikipedia page.

Presentation & Workshop Hosting — *Summer 2021*

- Conducted a presentation and workshop on **Django and the REST architectural style** for students at **Sharif University**. This session provided insights into the responsibilities of software engineers and **back-end developers** and offered an introduction to the basic concepts of Django.